

Referee Report

Nonparametric Identification of Demand without Exogenous Product Characteristics

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Summary

This paper shows that price counterfactuals in differentiated product demand models can be nonparametrically identified using “recentered instruments” formed from exogenous price shocks Z and possibly endogenous characteristics X . X need not be exogenous: it enters as a *proxy* for the unobserved utility index δ , and identification proceeds under a novel “faithfulness” condition replacing standard completeness.

Contribution relative to the literature: Berry and Haile (2014) established that nonparametric demand identification generally requires exogenous characteristic-based instruments alongside price instruments. Berry and Haile (2024) relaxed this with micro-data. This paper proves that neither exogenous characteristics nor micro-data are necessary for identifying *price counterfactuals*, and complements the recentered instrument framework of Borusyak and Hull (2023) and Borusyak et al. (2025).

Two caveats on framing. First, the “ J instead of $2J$ ” headline overstates the contrast with BH14, who never proved a sharp $2J$ lower bound. More importantly, the paper targets price counterfactuals—not the full inverse demand $\sigma^{-1}(\cdot)$ —a narrower object. The paper acknowledges this (pp. 3–4, 10), but the abstract could be clearer. Second, the identification logic underlying faithfulness—that conditional mean-independence from Z implies no dependence on P —appears implicitly in Imbens and Newey (2009), Torgovitsky (2015), and D’Haultfoeuille and Février (2015), as the authors acknowledge (footnote 6, Appendix B.2.2). The novelty is the formalization, the comparison with completeness, and the deployment with endogenous characteristics—not the underlying logic.

The mathematical execution is extremely sharp, and treating X as a proxy rather than an instrument is clarifying. However, I am frustrated by the gap between the theory and what an applied researcher can use, and I have concerns about restrictions in the proofs that have economic bite the paper does not discuss.

Essential Points

1. **The Need for Simulations.** What should an empirical researcher do differently after reading this paper? Nonparametric IV is an ill-posed inverse problem, and the paper provides no finite-sample evidence.

Concrete step forward: I am aware that Borusyak et al. (2025) provide Monte Carlo simulations for the parametric estimator, but those simulations do not speak to the nonparametric identification result or the faithfulness condition specifically. I need a simulation that shows: (i) in a DGP where faithfulness holds, recentered instruments recover price counterfactuals; (ii) when faithfulness is violated (e.g., weak proxy X), they do not. A standard Random Coefficients Logit with endogenous X would suffice.

2. **Economic Content of Regularity Conditions.** Two restrictions in the sufficient conditions have economic bite that the paper does not discuss:

- (a) *Smoothness (Proposition A.5):* Proving faithfulness under price separability requires restricting candidate inverse demand functions to be continuously differentiable with Lipschitz derivatives. Footnote 25 acknowledges this as a “complication,” but the implication is substantive: it shrinks the identified set Θ_I to smooth candidates only. Which economically relevant inverse demand functions are excluded? This belongs in the main text of Section 3.1.1, not a footnote.
- (b) *Limited Heteroskedasticity (Assumption 7(3)):* The authors commendably state this in the main text (p. 18). But what does this mean economically? The bound $\Sigma(x) \approx \Sigma_0$ ensures the Neumann series in Theorem A.1 converges, meaning identification requires the variance of $\delta \mid X$ to be approximately homoskedastic. In markets where product quality varies widely, this may fail. Which empirically relevant DGPs satisfy this bound?

3. **Faithfulness and the Weak Proxy Problem.** Faithfulness requires X to be a strong proxy for δ . To make this concrete, consider $H(\delta, P) = \delta + \beta P$. Under Assumption 3:

$$\mathbb{E}[\delta + \beta P \mid X, Z] = \mathbb{E}[\delta \mid X] + \beta \mathbb{E}[P \mid X, Z]. \quad (1)$$

Rejecting a false $\beta \neq 0$ requires $\text{Var}(\mathbb{E}[\delta \mid X])$ to be large enough that $\beta \mathbb{E}[P \mid X, Z]$ cannot offset it. If X is a weak proxy, this fails and researchers cannot reject incorrect demand specifications.

Concrete step forward: What happens to the identified set as $\text{Corr}(X, \delta) \rightarrow 0$? I encourage the authors to propose a “proxy strength” diagnostic analogous to the effective F-statistic (Montiel Olea and Pflueger, 2013).

Suggestions (Free Disposal)

1. **Assumption 3 ($Z \perp\!\!\!\perp \delta \mid X$) needs a formal DGP.** If firms choose X strategically, it acts as a collider between firm strategy and market unobservables, and conditional exogeneity of Z becomes hard to justify. A sequential DGP in Section 2 justifying Assumption 3—and stating its economic limitations—is needed.

2. **Move the DAG (Appendix B.1) to the main text.** It immediately clarifies the causal structure and the distinction between completeness and faithfulness.
3. **Delineate the standalone contribution.** The recentering framework is from Borusyak and Hull (2023); parametric estimation from Borusyak et al. (2025). Three of four authors overlap. A “roadmap” paragraph stating what is new here—independent of the companion papers—matters.
4. **Clarify which welfare objects are identified.** The demand curve $p \mapsto \sigma(\delta, p)$ is point-identified (Lemma 2), so within-market consumer surplus changes are identified, as footnote 8 notes. But cross-market comparisons of absolute utility levels depend on the unknown T and are not. Section 4 should delineate this boundary explicitly.
5. **$\tilde{X}$ notation.** Applied researchers need to know which characteristics require the index exclusion restriction. Re-emphasize in Section 4 that X must be excluded from random coefficients, while $\tilde{X}$ need not be.
6. **Engage with Akerberg and Crawford (2009).** They handle endogenous characteristics parametrically. Which DGPs are covered nonparametrically here but not by AC2009?
7. **Proximal causal inference parallel.** Using X as proxy for unobserved δ with instrument Z is formally analogous to the proximal causal inference framework (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). The paper cites Miao et al. (2018) for completeness (p. 8) but does not draw the broader structural parallel. Doing so would orient readers outside IO.
8. **Berry and Haile (2024) framing.** Calling BH24 purely a “micro data” approach (p. 6) may overstate the data-requirement advantage of the present paper, since some of BH24’s results also apply with aggregate shares.
9. **Missing reference.** Chesher (2003) on identification in nonseparable triangular models uses instrument-independence arguments structurally similar to faithfulness. Also, Compiani (2022) is cited but deserves more substantive engagement in Section 4 given the close overlap.
10. **Conciseness.** Section 3.1 is dense. Knife-edge cases (e.g., Proposition 6, discrete support) could move to the appendix.

References

- Akerberg, Daniel A. and Gregory S. Crawford, “Estimating Price Elasticities in Differentiated Product Demand Models with Endogenous Characteristics,” Working Paper 2009.
- Berry, Steven and Philip Haile, “Identification in Differentiated Products Markets Using Market Level Data,” *Econometrica*, 2014, 82 (5), 1749–1797.

- **and** –, “Nonparametric Identification of Differentiated Products Demand Using Micro Data,” *Econometrica*, 2024, *92* (4), 1135–1162.
- Borusyak, Kirill and Peter Hull**, “Nonrandom Exposure to Exogenous Shocks,” *Econometrica*, 2023, *91* (6), 2155–2185.
- , **Mauricio Caceres Bravo, and Peter Hull**, “Estimating Demand with Recentered Instruments,” 2025. arXiv preprint arXiv:2504.04056.
- Chesher, Andrew**, “Identification in Nonseparable Models,” *Econometrica*, 2003, *71* (5), 1405–1441.
- Compiani, Giovanni**, “Market Counterfactuals and the Specification of Multiproduct Demand: A Nonparametric Approach,” *Quantitative Economics*, 2022, *13* (2), 545–591.
- D’Haultfœuille, Xavier and Philippe Février**, “Identification of Nonseparable Triangular Models with Discrete Instruments,” *Econometrica*, 2015, *83* (3), 1199–1210.
- Imbens, Guido W. and Whitney K. Newey**, “Identification and Estimation of Triangular Simultaneous Equations Models Without Additivity,” *Econometrica*, 2009, *77* (5), 1481–1512.
- Miao, Wang, Zhi Geng, and Eric J Tchetgen Tchetgen**, “Identifying Causal Effects with Proxy Variables of an Unmeasured Confounder,” *Biometrika*, 2018, *105* (4), 987–993.
- Olea, José Luis Montiel and Carolin Pflueger**, “A Robust Test for Weak Instruments,” *Journal of Business & Economic Statistics*, 2013, *31* (3), 358–369.
- Tchetgen, Eric J Tchetgen, Andrew Ying, Yifan Cui, Xu Shi, and Wang Miao**, “An Introduction to Proximal Causal Inference,” *Statistical Science*, 2024, *39* (3), 375–390.
- Torgovitsky, Alexander**, “Identification of Nonseparable Models Using Instruments with Small Support,” *Econometrica*, 2015, *83* (3), 1185–1197.